(Dual Basis)

$\mathrm{Ann}(W) \leq V^*$ $(W \leq V)$

$\dim(V) < \infty$

$\dim(\mathrm{Ann}(W)) = \dim(V/W) = k$

↳ basis: $\{\underline{v_1}+W, \cdots, \underline{v_k}+W\}$

↓

basis: $\{f_1, f_2, \cdots, f_k\}$

Then $\phi: \mathrm{Ann}(W) \xrightarrow{\cong} V/W$ are isomorphic,

where $\phi(f_i) := \underline{v_i}+W$, $\forall i = 1, \cdots, k$

Note: def. of $\phi$

involves basis of $\mathrm{Ann}(W)$ & basis of $V/W$

WT: Define $\Phi: \mathrm{Ann}(W) \xrightarrow{\cong} (V/W)^*$

without using any choice of bases

Def.: $\Phi: \mathrm{Ann}(W) \to (V/W)^*$, $(f \in \mathrm{Ann}(W) \subseteq V^*)$

$f \mapsto \bar{f}$

where $\bar{f}(\underline{v}+W) = f(\underline{v})$

$\left(f: V \to \mathbb{F} \text{ in } \mathrm{Ann}(W) \text{ satisfies } W \leq \ker(f) \Rightarrow \bar{f}: V/W \to \mathbb{F}\right)$

(quotient space note)

Prop 1: $\bar{\Phi}$ is a natural linear trans.,

i.e. $\bar{\Phi}(af+bg) = a\bar{\Phi}(f) + b\bar{\Phi}(g)$ ①

Proof: (WTS: $\bar{\Phi}(f+g) = \overline{f+g} : V/W \to \mathbb{F}$)

(→ WTS: $\bar{\Phi}(f+g)(\underline{v}+W) = (\bar{\Phi}(f)+\bar{\Phi}(g))(\underline{v}+W)$, $\forall \underline{v}+W \in V/W$)

LHS:

$\bar{\Phi}(f+g)(\underline{v}+W)$

$= \overline{(f+g)}(\underline{v}+W)$

$= (f+g)(\underline{v})$

$= f(\underline{v}) + g(\underline{v})$

RHS:

$(\bar{\Phi}(f)+\bar{\Phi}(g))(\underline{v}+W)$

$= (\bar{f}+\bar{g})(\underline{v}+W)$

$= \bar{f}(\underline{v}+W) + \bar{g}(\underline{v}+W)$

$\overset{①}{=} f(\underline{v}) + g(\underline{v})$

$\Rightarrow$ LHS = RHS

Prop 2: $\Phi: Ann(W) \to (V/W)^*$ is injective

Proof: Suppose $f \in \ker(\Phi)$ (Recall: $f: V \to \mathbb{F}$ is in $Ann(W)$)

Then $\Phi(f) = 0_{(V/W)^*}: V/W \to \mathbb{F}$,

i.e. $\Phi(f)(v+W) = 0_{(V/W)^*}(v+W)$, $\forall v+W \in V/W$

$\Rightarrow f(v+W) = 0_{\mathbb{F}}$

$\Rightarrow f(v) = 0$, $\forall v \in V$

$\Rightarrow f = 0_{V^*} = 0_{Ann(W)}$

Corollary: If $\dim(V) < \infty$,

then $\Phi : Ann(W) \xrightarrow{\cong} (V/W)^*$

is a natural isomorphism

Proof: 已知道: (1) $\dim(Ann(W)) = \dim((V/W)^*)$

(2) (Prop. 2 $\Rightarrow$) $\Phi$ is inj.

$\Rightarrow \Phi$ is bij.

Remark: The Corollary lies on

(1) $\dim(V) = \dim(W) < \infty$

(2) $T : V \to W$ is inj.

$\Rightarrow T : V \xrightarrow{\cong} W$

(This) fails if $\dim(V) = \infty$

need

(已经证有: $\Phi : Ann(W) \xrightarrow{\cong} (V/W)^*$ for $\infty$-dim. V. spaces)

Transpose Map

Def.: $T : V \to W$

the transpose of $T$ is $T^* : W^* \to V^*$

$f \in W^* \longmapsto T^*(f) \in V^*$

s.t. $\forall f \in W^*$, $\forall v \in V$,

$[T^*(f)](v) := f(\underbrace{T(v)}_{\in W}) \in \mathbb{F}$

Remark: (1) By def., $T^*(f): V \to \mathbb{F}$ is a map

But $T^*(f) = f \circ T \in V^*$ is a linear functional

(不同？⊛)

(2) By (1), $T^*$ does map $W^*$ to $V^*$

In fact, $T^*: W^* \to V^*$ is a linear trans.

e.g.: WTS: $T^*(f+g) = T^*(f) + T^*(g)$

$T^*(f+g) \in V^*$, $T^*(f), T^*(g) \in W^*$

↳ WTS: $T^*(f+g)(\underline{v}) = (T^*(f) + T^*(g))(\underline{v})$, $\forall \underline{v} \in V$

Proof: RHS $=$

$= T^*(f)(\underline{v}) + T^*(g)(\underline{v})$

$= f(T(\underline{v})) + g(T(\underline{v}))$

$= (f+g)(T(\underline{v}))$

LHS $=$ ~

$=$

Prop.1: $T: V \to W$

1. $T$ is inj. $\Rightarrow T^*$ is surj.

2. $T$ is surj. $\Rightarrow T^*$ is inj.

⛤ Proof of 2.:

Suppose $T: V \to W$ is surj.

$\Rightarrow T^*: W^* \to V^*$

and suppose $f \in \ker(T^*)$

$\Rightarrow T^*(f) = \underline{0}_{V^*}$

$\Rightarrow T^*(f)(\underline{v}) = \underline{0}_{V^*}(\underline{v})$, $\forall \underline{v} \in V$

$\Rightarrow f(T(\underline{v})) = 0$

$T$ is surj. $\Rightarrow f(\underline{w}) = 0$, $\forall \underline{w} = T(\underline{v}) \in W$ ⛤

$\Rightarrow f = \underline{0}_{W^*}$

Prop 2: (见图: Prop. 5.6)

$$\left((T_{BA})^t = T^*_{A^*B^*},\ T: \underset{(A)}{V} \to \underset{(B)}{W},\ T^*: \underset{(B^*)}{W^*} \to \underset{(A^*)}{V^*}\right)$$

Proof: 见本周 Tut ~~#~~

Recall: $\Phi : Ann(W) \to (V/W)^*$

Recall: For $W \leq V$,

the "cananical projection" $\pi_W : V \to V/W$,

$\pi_W(\underline{v}) = \underline{v} + W$ $\quad (\ker(\pi_W) = W)$

Construct the inverse of $\Phi$ using transpose

Consider $(\pi_W)^* : (V/W)^* \to V^*$

Then $\forall f \in (V/W)^*$,

$(\pi_W)^*(f) \in V^*$ is given by

$(\pi_W)^*(f)(\underline{v}) := f(\pi_W(\underline{v})) := f(\underline{v}+W)$, $\forall \underline{v} \in V$

Then $\forall \underline{w} \in W$, $(\pi_W)^*(f)(\underline{w}) = f(\underline{w}+W)$

$\underline{\textcircled{?}} = f(\underline{0}+W)$

$= f(\underline{0}_{V/W}) = 0$

$\Rightarrow (\pi_W)^*(f) \in Ann(W)$

$\Rightarrow (\pi_W)^* : (V/W)^* \to Ann(W)$

$\textcircled{?} \Rightarrow \downarrow$ is the inverse of $\Phi$